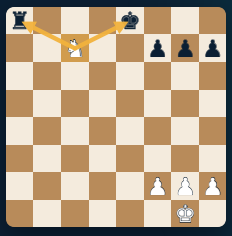
TURNING A CHESS ENGINE INTO SCIENCE AI

Chess AI is now solving science problems.

The search that lets a computer beat chess champions now helps scientists at **Brookhaven National Laboratory** decide which measurement to take next. In one week, students build a chess engine named **Remus** in Python, then turn its thinking on a science puzzle.

5 days, 8:30 to 4:30 | All students welcome | Coding and chess

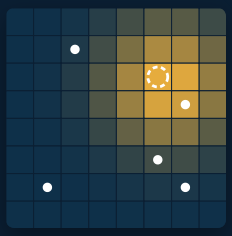


ON THE BOARD

Which move next?

Remus looks ahead and finds the knight fork.

SAME SEARCH: LOOK AHEAD, SCORE, CHOOSE



IN THE LAB

Which measurement next?

An AI picks the spot most likely to reveal the hidden peak.

Five days, four engines EACH DAY ADDS A VERSION

DAY 1

Remus1 plays legal chess

Every rule of chess. By the afternoon Remus1 plays you in Lucas Chess, at random.

`193 lines of code`

DAY 2

Remus2 searches ahead

Scoring and alpha-beta search. Remus2 plays Remus1, then you.

`161 lines of code`

DAY 3

Remus3 adds the biggest gains

An opening book, a memory of positions, and the clock. Remus3 plays Remus2.

`193 lines of code`

DAY 4

Remus4 adds everything else

Draws, pruning, and the endgame. Remus4 plays Remus3 and Remus4-Rust.

`159 lines of code`

DAY 5

Solve a science problem

Find a hidden peak, name the crystals in an X-ray pattern, and share your results.

`57 lines of code`

Every day FIVE WEEKDAYS IN A ROW

8:30 to 10:00	10:00	10:30 to 12:00	12:00 to 1:00	1:00 to 2:30	2:30	3:00 to 4:30
Period 1	Break	Period 2	Lunch	Period 3	Break	Period 4

How a period runs: a short talk on the next chess rule or engine idea, then students type the next part of Remus from the instructor's version and run it. One types, one reads aloud and checks, and you swap every period.

YOUR SCIENCE CHALLENGE ON DAY 5

Find the hidden peak.

Program an AI that tracks down a hidden hot spot in as few measurements as possible. Then aim your chess search at a real beamline question: which crystals are in a powder? Scientists at Brookhaven's NSLS-II ask it every day.

You take home your four engines and science programs on a USB drive, and you can install Lucas Chess free at home on Windows to keep playing against the Remus engine you have developed.

For parents

Does a student need a laptop?

No. Each student pair uses a class laptop with everything installed. Twenty students work in ten student pairs with their teacher, coding and playing chess together.

Will a student need online accounts?

No. There are no sign-ups of any kind. All work stays on the class laptops and goes home on USB.

Does a student need to know how to code?

Yes, the basics: functions, loops, and lists, in any language such as Java. Students also need to know how to play chess, including notation such as e4 and Nf3.

You are ready if you

- ✔ know some coding
- ✔ can stay focused while tracing algorithms
- ✔ know how to play chess, including notation
- ✔ want to learn how AI works